

CHAPTER 9

RIPARIAN AND LITTORAL BOUNDARIES

9.1 INTRODUCTION

Water boundaries are possibly the most ancient and widely used of humankind's boundaries. Today this area of surveying and law aspects are probably the least understood by the courts and the practicing surveyors. Surveyors in a few states are concerned with *tidal boundaries as well as non-tidal boundaries, but have to worry about title boundaries*. Throughout recorded history, societies have recognized water boundaries as the limits of their territories. As an early example of this, very detailed laws regarding water boundaries were included in the Roman Civil Code known as the Institutes of Justinian, prepared under the direction of Emperor Justinian in A.D. 529. That document serves as the root of today's water boundary procedures. The following translations¹ of pertinent excerpts from that code illustrate the detail and the similarity of that code to modern water law. As may be seen, that code addresses many of the same issues pertinent in today's society, including issues such as public ownership of navigable waters, the location of the boundary between publicly owned waters and privately owned littoral or riparian land, public access to navigable waters, riparian rights to newly formed lands, and ownership of newly formed islands.

The U.S. Constitution enters into the legal and historical when it enacted the *Interstate Commerce Clause* and its resultant application to anything that applied to interstate commerce, and declaring all navigable rivers navigable.

Section 1. By the law of nature these things are common to mankind—the air, running water, the sea, and consequently the shores of the sea. No one, therefore, is forbidden to approach the sea-shore, provided that he respects habitations, monuments, and buildings, which are not, like the sea, subject only to the law of nations. . . . The sea-shore, that is the shore as far as the waves go at furthest, was considered to belong to all men.

Section 2. All rivers and ports are public; hence the right of fishing in a port or in rivers, is common to all men.

Section 3. The sea-shore extends to the limit reached by the greatest winter flood.

Section 4. The public use of the banks of a river is part of the law of nations, just as is that of the river itself. All persons, therefore, are as much at liberty to bring their vessels to the bank, to fasten ropes to the trees growing there, and to place any part of their cargo there, as to navigate the river itself. But the banks of a river are the property of those whose land they adjoin; and consequently the trees growing on them are also the property of the same persons.

Section 5. The public use of the sea-shore, too, is part of the law of nations, as is that of the sea itself; and therefore any person is at liberty to place on it a cottage, to which he may retreat, or to dry his nets there, and haul them from the sea; for the shores may be said to be the property of no man, but are subject to the same law as the sea itself, and the ground or sand beneath it.

Section 20. Moreover, the alluvial soil added by a river to your land becomes yours by the law of nations. Alluvion is an imperceptible increase; and that is added by alluvion, which is added so gradually that no one can perceive how much is added at any one moment of time.

Section 21. But if the violence of a river should bear away a portion of your land and unite it to that of your neighbour, it undoubtedly still continues yours. If, however, it remains for a long time united to your neighbour's land, and the trees, which it swept away with it, take root in his ground, those trees from that time become part of your neighbour's estate.

Section 22. When an island is formed in the sea, which rarely happens, it is the property of the first occupant; for before the occupation it belongs to no one. But when an island is formed in a river, which frequently happens, then it occupies the middle of the river, it belongs respectively to those who possess the lands near the banks on each side of the river, in portion to the extent along the banks of each man's estate. But if the island is nearer to one side than the other, it belongs to those persons only who possess lands contiguous to the bank on that side. But if a river divides itself at a certain point, and lower down unites again, thus giving to any one's land the form of an island, the land still continues to belong to the person to whom it belonged before.

Section 23. If a river, entirely forsaking its natural channel, begins to flow in another direction, the old bed of the river belongs to those who possess the lands adjoining its banks, in proportion to the extent along the banks of their respective estates. The new bed follows the condition of the river, that is, it becomes public. And, if after some time the river returns to its former channel, the new bed again becomes the property of those who possess the lands along its banks.

Section 24. The case is quite different if any one's land is completely inundated; for the inundation does not alter the nature of the land, and therefore, if the water recedes, the land remains indisputably the property of the same owner.

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As may be seen from these excerpts from the Institutes of Justinian, water boundaries were complex even seven centuries ago. Today, water boundaries continue

to be possibly the most complex of man's boundaries. The complexity of these boundaries is due to the fact that both the level of the water and the shoreline itself are constantly changing. This complexity resulted in the development of unique principles, such as those illustrated in the excerpts from the Institutes of Justinian, to guide the determination of water boundaries. The currently prevailing versions of those principles are the subject of this chapter. The current principles gleaned from historical and current case law include the following:

- PRINCIPLE 1. The high seas are considered as a global commons accessible to all. A marginal territorial sea along the coastline is subject to exclusive claim by each coastal nation and/or state.
- PRINCIPLE 2. Although the territorial sea of the United States currently extends 12 nautical miles seaward of the mean lower low-water line, each coastal state has ownership of the area out to 3 geographical miles, except for Florida and Texas, whose boundaries extend out to 3 marine leagues (9 miles) in the Gulf of Mexico.
- PRINCIPLE 3. With statehood, all coastal states received ownership of all natural submerged lands affected by the tides (tidelands) within their boundaries, regardless of their navigability. Some states have relinquished a claim to those tidelands that are not navigable-in-fact.
- PRINCIPLE 4. In most of the U.S. coastal states, the landward boundary between the state-owned submerged lands in tidal waters and bordering uplands is the mean high-water line.
- PRINCIPLE 5. Exceptions to the use of the mean high-water line as the sovereign/upland coastal boundary include several northeastern states that recognize the mean low-water line as that boundary. In addition, exceptions exist in some areas where land title has roots in a grant from a sovereign power where civil law, as opposed to Anglo-American common law, prevailed. By the civil law, the shore extends as far "as the highest waves reach in winter" as opposed to the mean high-water line of the common law.
- PRINCIPLE 6. The submerged lands under natural navigable nontidal waters are owned by the public in most of the United States, although some states do not claim such ownership.
- PRINCIPLE 7. The boundary of publicly owned nontidal submerged lands is the ordinary high-water mark for many states, although a number of states claim only to the ordinary low-water mark.
- PRINCIPLE 8. A meander line of the Public Land Survey generally does not constitute a boundary. Rather, it represents the approximate location of the shoreline at a certain point in time.
- PRINCIPLE 9. Submerged lands under non-publicly owned waters are customarily owned by the adjoining owners, with the ownership extending to the center of the stream.

- PRINCIPLE 10. Swamp and overflowed lands are not part of the public trust submerged lands. In the public land states, such lands were granted to the states for subsequent improvement and conveyance to private owners.
- PRINCIPLE 11. The United States has authority to regulate activities on all navigable waters of the country.
- PRINCIPLE 12. Riparian and littoral property is often subject to public regulation, which may create regulatory limit lines, which, although different from property lines, may also affect the utility of the property.
- PRINCIPLE 13. The general rule is that water boundaries change with gradual and natural changes to the shoreline, while those boundaries do not change with sudden or human-made changes. Land that is gradually and naturally built up by accretion or exposed by reliction accrues to the riparian owner. Land that is gradually and naturally eroded away or permanently inundated by the water accrues to the owner of the bed.
- PRINCIPLE 14. Regarding the apportionment of riparian and littoral rights:
- The objective of apportionment is to distribute either newly formed land or rights in water in proportion to the length of water frontage. Therefore, the length of the affected tract's water frontage, not its total area or upland lateral lot lines, generally controls the apportionment of riparian rights.
 - For the division of newly formed land, the point of departure for a division line is the intersection of the upland lateral boundary and the shoreline at the time of the subdivision of the area. For the division of water area, the point of departure for a division line is the intersection of the upland lateral boundary and the current shoreline.
 - For the division of newly formed land, the newly formed shoreline should be proportioned relative to the respective lengths of the old and new shoreline.
 - In streams, the general rule for the division of water area is to project the division line in a direction perpendicular to the thread of the stream.
 - In large bays, oceans, or wide rivers where a thread is not readily discernible, the general rule for the division of water area is to run lines perpendicular to various other baselines such as the shoreline if it is relatively straight, the bulkhead or pier line, or the channel line.
 - In lakes, the general rule for the division of water area for round lakes is for side lines to be drawn to a center point, creating pie slice-shaped areas. For long lakes, it is generally held that a centerline should be constructed and lines projected out perpendicular to the centerline.
 - In partially enclosed areas, such as coves, the general rule for the division of water area is to proportion the closing line across the mouth of the area based on the respective lengths of the shoreline.
- PRINCIPLE 15. Emergent islands are generally considered to belong to the owner of the bed of the water body. In public land states, omitted lands

(such as islands existing at the time of, but not surveyed in, the public land survey) remain in the ownership of the federal government until specifically conveyed. Islands emerging after the conveyance of the water body to the states with statehood, belong to the owner of the bed of the water body.

9.2 OWNERSHIP OF THE SEAS²

Principle 1. *The high seas are considered as global commons accessible to all. A marginal territorial sea along the coastline is subject to exclusive claim by each coastal nation and/or state.*

As illustrated by the excerpts from the Roman Institutes of Justinian cited in the introduction to this chapter, the high seas have long been considered as a global commons, open and available to all. Yet, as navigation of the seas became more common, coastal nations began reserving the portion of the seas bordering their nations as their exclusive territory. This concept of a marginal or territorial sea evolved as a matter of self-defense on the part of the coastal nations. That zone was, for practical purposes, the zone which the nation could comfortably defend for its exclusive use from hostile shipping. By the eighteenth century, this was more or less standardized by the so-called cannon shot rule. By that informal rule, the boundary of the marginal sea was the distance from shore that a cannon shot could typically reach. That distance was generally considered to be 3 miles.

More recently, as various nations began exploiting the riches of the sea and sea floor, widely ranging claims have been made, which vary considerably from the “cannon shot” rule. This issue has been frequently debated at international conferences on the law of the sea. In 1982, a United Nations conference adopted an international treaty that has somewhat standardized maritime boundaries. That treaty gave coastal nations a territorial sea extending out to 12 miles from shore, with exclusive economic zones of 200 miles or to a maximum of 350 miles if the country’s continental shelf extends out that far. Bays and partially enclosed coastal indentations are considered to be exclusive territory of the surrounding nation under this treaty. As with earlier law-of-the-sea treaties, the latest treaty also stresses that the open seas are common to all and not controllable by any individual nation. Although not a signatory to the United Nations Treaty of the Sea, the United States has adopted similar claims to its adjacent waters.

9.3 OWNERSHIP OF THE U.S. TERRITORIAL SEA

Principle 2. *Although the territorial sea of the United States currently extends 12 nautical miles seaward of the mean lower low-water line, each coastal state has ownership of the area out to 3 geographical miles, except for Florida and Texas, whose boundaries extend out to 3 marine leagues (9 miles) in the Gulf of Mexico.*

Federal Claims

Prior to the 1930s, the United States claimed a 3-mile-wide territorial sea and made no attempt to exert jurisdiction beyond that limit. During the 1930s, the jurisdiction of the United States for law enforcement was extended to the limits of a 12-mile “contiguous zone” in connection with enforcement of prohibition laws (rumrunners had been able to easily escape interception with the 3-mile territorial jurisdiction. That jurisdictional extension was of an extra-territorial nature and did not extend the territorial sea ownership.

In 1945, with the development of offshore oil drilling, the claims of the United States were further extended by a presidential proclamation stating, “. . . the Government of the United States regards the natural resources of the subsoil and seabed of the continental shelf beneath the high seas but contiguous to the coasts of the United States as appertaining to the United States, subject to its jurisdiction and control.”³ The continental shelf is a gently sloping plain of land along the coasts of most islands and continents, and it varies in width, from a few miles to hundreds of miles. The continental shelf is considered to end where the continental slope begins to drop more steeply to the ocean floor.

In 1977, the United States claimed exclusive fishing management jurisdiction to the 200-mile limit, regardless of whether this line went beyond the continental shelf. In 1982, the previously mentioned United Nations Law of the Sea conference adopted a treaty establishing a 12-mile territorial sea, and, in 1988, the United States adopted an equivalent-width territorial sea. This occurred when President Ronald Reagan extended the U.S. territorial sea “. . . to the limits permitted by international law . . .” with a presidential proclamation. That document proclaimed as follows:

The territorial sea of the United States henceforth extends to 12 nautical miles from the baseline of the United States determined in accordance with international law.⁴

Therefore, the current policy of the United States is to claim ownership of a territorial sea of 12 nautical miles from its coastline, to claim exclusive jurisdiction over the natural resources to the extent of the continental shelf, and to claim exclusive fishing rights to 200 miles. The coastline, from which the width of the territorial sea is measured, is considered to be the low-water line, as depicted on a nation’s nautical charts. For the United States, that line is the mean lower low-water line of either the mainland or any island (any land mass, including rocks, with surfaces above the mean high-water line) lying within the nation’s territory.

State Claims

Until the 1930s, it was generally accepted that the public of each of the coastal states, rather than the federal government, owned the marginal sea off its coast as well as the interior navigable waters. However, with the discovery of oil in the submerged lands off the California coast, disputes arose as to the ownership of such lands. The resulting conflict led to a series of court cases that held that the federal government,

rather than the states, was owner of the marginal sea. Following those decisions, the U.S. Congress passed the Submerged Lands Act (43 U.S.C., s1301-1 1970) in 1953, which conveyed a band of submerged lands bordering the coastal states to the public of those states. That act defines the seaward limit of the public trust waters of each of the coastal states as follows:

Section 4—The seaward boundary of each original coastal State is hereby approved and confirmed as a line three geographical miles distant from its coastline or, in the case of the Great Lakes, to the international boundary. Any State admitted subsequent to the formation of the Union which has not already done so may extend its seaward boundaries to a line three geographical miles distant from its coastline or to the International boundaries of the United States in the Great Lakes or any other body of water traversed by such boundaries. . . . Nothing in this section is to be construed as questioning or in any manner prejudicing the existence of any State's seaward boundary beyond three geographical miles if it was so provided by its constitution or laws prior to or at the time such State became a member of the Union, or if it has been heretofore approved by Congress. (Note: A geographical mile is the length of 1 minute of arc on the equator or 6087.09 feet on the Clark Spheroid of 1866. A marine league is three geographical miles.⁵)

The question of whether any of the states were entitled, under the act, to submerged lands greater than 3 geographical miles from the coastline was decided by the Supreme Court in 1960.⁶ That decision held that Texas and Florida were entitled to submerged lands extending 3 leagues into the Gulf of Mexico due to the extent of their boundaries at the time of admission into the Union. The same decision held that Louisiana, Mississippi, and Alabama were entitled to a marginal sea of only 3 geographical miles.⁷

As a result, the public trust lands of each of the coastal states are now considered to be the navigable interior waters of their state together with the portion of the territorial sea out to 3 miles from the mean lower low-water line (even though the current policy to the federal government is to claim ownership out to 12 miles), except for the states of Florida and Texas, with lands extending out to 3 leagues into the Gulf of Mexico.

9.4 OWNERSHIP OF INTERIOR TIDAL WATERS OF THE UNITED STATES

Principle 3. *With statehood, all coastal states received ownership of all natural submerged lands affected by the tides (tidelands) within their boundaries, regardless of their navigability. Some states have relinquished a claim to those tidelands which are not navigable-in-fact.*

All natural tidal waters within their boundaries, unless validly conveyed, are considered to have been conveyed to the public of each of the U.S. coastal states. This is the case whether or not such waters are navigable. Therefore, unless a state has relinquished its claim, such waters are considered to be owned by the public of each state. This principle was confirmed in a case in the State of Mississippi,⁸ in which both the chancery court and the Mississippi Supreme Court ruled that as long

as the tide ebbed and flowed in a natural water body, it was publicly owned. Salient excerpts from that colorful court opinion are as follows:

The early federal cases refer to the trust as including all lands within the ebb and flow of the tide.

... it is our view that as a matter of federal law, the United States granted to this State in 1817 all lands subject to the ebb and flow of the tide and up to the mean high water level, without regard to navigability.

Yet so long as by unbroken water course—when the level of the waters is at mean high water mark—one may hoist a sail upon a toothpick and without interruption navigate from the navigable channel/area to land, always afloat, the waters traversed and the lands beneath them are within the inland boundaries we consider the United States set for the properties granted the state in trust.

The case was appealed to the U.S. Supreme Court,⁹ which concurred with the state court in ruling that all coastal states received all lands over which tidal waters flow, and that Mississippi still does. The opinion noted that this ruling “will not upset titles in all coastal states” since it “does nothing to change ownership rights in states which previously relinquished a public trust claim to tidelands such as those at issue here.” Thus, it is presumed that in states with well-established case law stating that only navigable tidal waters were publicly owned, navigability-in-fact would be the salient criterion for public ownership. However, that issue has yet to be judicially tested.

9.5 LANDWARD BOUNDARY OF TIDAL WATERS

Principle 4. *In most of the U.S. coastal states, the landward boundary between the state-owned submerged lands in tidal waters and bordering uplands is the mean high-water line.*

Principle 5. *Exceptions to the use of the mean high-water line as the sovereign/upland coastal boundary include several northeastern states that recognize the mean low-water line as that boundary. In addition, exceptions exist in some areas where land title has roots in a grant from a sovereign power where civil law, as opposed to Anglo-American common law, prevailed. By the civil law, the shore extends as far “as the highest waves reach in winter” as opposed to the mean high-water line of the common law.*

Under English common law, submerged lands underlying tidal waters belonged to the Crown. In the United States, the public of each state is sovereign and thus holds title to the beds under navigable waters. That concept was first stated in the U.S. Supreme Court case of *Martin v. Waddell*¹⁰ in 1842. According to the Court in that case,

When the revolution took place the people of each state became themselves sovereign and in that character hold the absolute right to all their navigable waters in the soils under them for their own common use. . . .

General Anglo-American Common Law

In most of the United States, the shoreward boundary of such public trust waters is defined by Anglo-American common law. Possibly the earliest definition of the boundary was provided by Thomas Digges, an engineer, surveyor, and lawyer during the reign of Queen Elizabeth I, in a book entitled *Proofs of the Queen's Interest in Land Left by the Sea and the Salt Shores Thereof*. Digges suggested that the queen's ownership of the waters included the foreshore, which is the area between high and low tides. In the following century, Lord Matthew Hale, a jurist who was to become the British chief justice, built on that concept in his treatise *De Jure Maris*,¹¹ written about 1666. In that writing, Hale posited that the foreshore, which is overflowed by "ordinary tides or neap tides, which happen between the full and change of the moon," belonged to the Crown. In 1854, this definition was clarified in English common law by the case of *Attorney General v. Chambers*.¹² The *Chambers* case ruled that the ordinary high-water mark was to be found by "the average of the medium tides in each quarter of a lunar evolution during the year (which line) gives the limit, in the absence of all usage, to the rights of the Crown on the seashore."

In the United States, the boundary was further clarified in 1935 with the U.S. Supreme Court's landmark decision in *Borax Consolidated v. City of Los Angeles*,¹³ as follows:

In view of the definition of the mean high tide, as given by the United States Coast and Geodetic Survey that "mean high water at any place is the average height of all the high waters at that place over a considerable period of time", and the further observation that "from theoretical considerations of an astronomic character" there should be "a periodic variation in the rise of water above sea level having a period of 18.6 years", the Court of Appeals directed that in order to ascertain the mean high tide line with requisite certainty in fixing the boundary of valuable tidelands, such as those here in question appear to be, "an average of 18.6 years should be determined as near as possible". We find no error in that instruction.

As may be seen, the *Borax* decision applied modern technical knowledge to set forth a specific technique for precisely locating the boundary in question by taking into account all of the periods of the astronomical forces affecting the tides to obtain an average height. Thus, the mean high-water line represents an attempt to define the boundary between publicly owned submerged lands and uplands subject to private ownership as the average upper reach of the daily tides. The result is a line that is exceeded by the high tide on approximately one-half of the tidal cycles.

Case law in most of the coastal states has generally followed the English common law and its updated definition as put forth in the *Borax* decision. Some states have codified their statutory law on this subject. As an example, in Florida, the Coastal Mapping Act of 1974 (Chapter 177, Part II, Florida Statutes) declares that "mean high water line along the shores of lands immediately bordering on navigable waters is recognized and declared to be the boundary between the foreshore owned by the State in its sovereign capacity and upland subject to private ownership." That statute further defines the mean high-water line using the *Borax* concept.

Exceptions Based on Massachusetts Colonial Ordinance

Several northeastern states are exceptions to those using the mean high-water line as a coastal boundary. Those states recognize the mean low-water line as the sovereign/upland boundary. The basis for that ownership is generally considered to be an early Massachusetts colonial ordinance, which was enacted in 1641 and revised in 1647. Thus, it is possibly the earliest reference to water boundaries in the New World. The ordinance provided as follows:

. . . in all creeks, coves and other places, about and upon salt water where the Sea ebs and flows, the Proprietor of the land adjoining shall have proprietie to the low water mark where the Sea doth not ebb above a hundred rods, and not more wheresoever it ebs farther.¹⁴

The language of the ordinance has generally been interpreted as defining the waterward boundary of lands bordering on tidal waters as the mean low-water line unless that line is 100 rods distant from the mean high-water line. If there is more than 100 rods' distance between the two lines, then the boundary is a line 100 rods seaward of the mean high-water line. Therefore, ownership of riparian or littoral lands under this legal system includes the tidal flats often found in such areas. Much of the ordinance remains in effect in the Commonwealth of Massachusetts and the state of Maine, which was separated from Massachusetts in 1820, as part of the common law. Other states rely in parts of the ordinance as part of their statute law or common law.

Exceptions Based on Civil Law

Other exceptions to the use of the mean high-water line as a coastal boundary exist where civil law, as opposed to Anglo-American common law, controls. This is generally in areas where the land title has its roots in a grant from a sovereign power where civil law, such as the Roman Institutes of Justinian, prevailed. Possibly because the Roman civil law code was developed in an area of the Mediterranean with little daily tidal range, such law does not define the coastal boundary in terms of daily tide but, rather, in terms of seasonal water level changes. As an example, a translation of a portion of the Institutes of Justinian follows:

The sea-shore, that is, the shore as far as the waves go at furthest, was considered to belong to all men. . . . The sea shore extends as far as the greatest winter floods runs up.¹⁵

That difference between the common law and civil law boundary definitions was clearly stated in the previously discussed *Borax* decision as follows:

By the civil law, the shore extends as far as the highest waves reach in winter. But by the common law, the shore "is confined to the flux and reflux of the sea at ordinary tides."

As an example of the use of the civil law definition, Louisiana has adopted the line of the highest winter tide as its coastal boundary. Interestingly, the highest

winter tide in the northern Gulf of Mexico is usually *less* than the highest tide in other parts of the year because of the prevailing annual constituent of the tides in the Gulf. The State of Hawaii also appears to follow the common-law approach with a coastal boundary defined as “the upper reach of the wash of the waves.”¹⁶ Another example is the State of Texas, which has recognized the civil law definition in areas of the state with origins of land title in Spanish or Mexican land grants. The Commonwealth of Puerto Rico also follows the civil law. For the latter two jurisdictions, it has been held that the limit of ownership is controlled by old Spanish law contained in *Las Siete Partidas*,¹⁷ written in the thirteenth century, where the boundary is defined as follows:

... e todo aquel lugar es llamado ribera de la mar quanto se cubre el agua della, quanto mas crece en todo el ano, quier en tiempo del invierno o verano. . . . (Partida 3, Title 28, Law 4)

That definition appears to use the highest reach of the water whether in winter or summer (“*invierno o verano*”). The supreme court of Texas has interpreted that definition as being the mean higher high-water line in one case,¹⁸ although, in coastal Texas, the mean higher high-water line is almost identical in height to the mean high-water line due to the prevailing diurnal tide in that area. In Puerto Rico, the boundary is usually determined by use of vegetative and geological features reflecting higher water levels. Tests of that approach indicate that such a line is generally higher than the highest astronomic tide and therefore reflects a line created by higher water levels such as those associated with storm events.¹⁹

9.6 OWNERSHIP OF NONTIDAL NAVIGABLE WATERS

Principle 6. *The submerged lands under natural navigable nontidal waters are owned by the public in most of the United States, although some states do not claim such ownership.*

The public trust doctrine also extends to navigable nontidal waters in most of the United States. This is actually a long-established concept addressed in the Roman Civil Code of Emperor Justinian I, which expressly declared all rivers as well as the seas to be a commons area. However, early English common law considered only tidal waters to be sovereign, possibly due to the fact that the English island has few inland waters with a capacity for public navigation. The English common-law position is illustrated by the following:

That rivers not navigable (that is, freshwaters rivers of what kind so ever) do, of common right belong to the owners of the soil adjacent. But that rivers, where the tide ebbs and flows, belong to the State or public.²⁰

In the United States, case law has generally differed from English common law on this issue. For example, the Massachusetts Colonial Ordinance of 1641 held that great ponds, in excess of 10 acres, could not be conveyed to private ownership. In

1876, this issue was addressed by the U.S. Supreme Court with the case of *Barney v. Keokuk*,²¹ as follows:

The confusion of navigable with tide water, found in the monuments of the common law, long prevailed in this country, notwithstanding the broad differences existing between the extent and topography of the British island and that of the American continent. . . . And since this court . . . has declared that the Great Lakes and other navigable waters of the country, are, in the strictest sense, entitled to the denomination of navigable waters, and amenable to the admiralty jurisdiction, there seems to be no sound reason for adhering to the old rule as to the proprietorship of the beds and shores of such waters. It properly belongs to the States by their inherent sovereignty. . . .

State common law has generally followed this lead. As a result, many of the states, although there are exceptions, claim title to the beds of navigable waters regardless of whether the waters are tidally affected. Navigability of nontidal waters for title purposes is generally a question of navigability-in-fact, with many states adopting an early test suggested by federal case law,²² as follows:

and they are navigable in fact when they are used or susceptible of being used in their ordinary condition, as highways for commerce, over which trade and travel are or may be conducted in the customary modes of trade and travel over water.

9.7 LANDWARD BOUNDARIES OF NONTIDAL WATERS

Principle 7. *The boundary of publicly owned nontidal submerged lands is the ordinary high-water mark for many states, although a number of states claim only to the ordinary low-water mark.*

There was no precedent in English common law regarding boundaries of nontidal waters due to the fact that early English common law did not consider nontidal waters to be privately owned. Further, with the lack of a predictable rise and fall of water level such as that observed in tidal waters, definitions used in tidal waters could not be used directly. Therefore, American case law adopted the physical fact test to determine the equivalent of mean high water in nontidal waters. To distinguish that boundary from a mathematically derived tidal boundary (mean high water), that boundary is generally called the *ordinary high-water mark*. Prior to the *Borax* decision, the term *ordinary high-water mark* was also frequently used to describe what is now commonly called the *mean high-water line* in tidal waters. (It should be noted that a number of states use the *ordinary low-water mark* for the boundary of nontidal waters.)

The leading definition in federal case law, *Howard v. Ingersol*,²³ gives the following instructions for determining the ordinary high-water mark:

This line is to be found by examining the bed and banks and ascertaining where the presence and action of waters are so common and usual and so long continued in all ordinary years, as to mark upon the soil of the bed a character distinct from that of the banks, in respect to vegetation, as well as in respect to the nature of the soil itself.

As may be seen, that opinion suggests the use of physical evidence of long-standing water to determine the boundary, and case law in most of the states has adopted this approach. Further, surveying practice has followed in that direction as suggested by the following guideline from the *Manual of Surveying Instruction*.²⁴

The most reliable indicator of mean (ordinary) high water elevation is the evidence made by the water's action at its various stages, which are generally well marked in the soil. In timbered localities, a very certain indication of the locus of the various important water levels is found in the belting of the native forest species.

Considering both the legal and technical guidance, evidence that should be evaluated for determining the appropriate location of the ordinary high-water mark includes the following:²⁵

1. *Geomorphological features*: These include features indicative of the natural limits of water bodies such as escarpments and natural levees.
2. *Vegetation*: This category includes evidence such as the lower limit of terrestrial vegetation and areas where vegetation has been wrested away by wave action or stream currents.
3. *Change in the character of the soil*: This includes evidence such as differences in organic content due to leaching and the landward limit of stratified beach deposits where waterward transport of eroded shoreline has resulted in a natural sorting of material.
4. *Hydrological records*: Although earlier case law has typically rejected the use of statistical averaging of water level data, some recent cases have suggested that hydrological data are potentially a valuable class of evidence. Further, tests of one proposed method for analyzing hydrographic data shows agreement with traditional physical evidence.²⁶

Some states have adopted the ordinary low-water mark as the boundary of navigable nontidal waters. Such a boundary is difficult to locate due to soil and vegetative evidence generally not existing at lower water stands. Typically, the best evidence of the ordinary low-water line includes escarpments associated with low-water stands and long-term hydrological records.

9.8 SIGNIFICANCE OF PUBLIC LAND SURVEY MEANDER LINES²⁷

Principle 8. *A meander line of the Public Land Survey generally does not constitute a boundary. Rather, it represents the approximate location of the shoreline at a certain point in time.*

The meandering of navigable water bodies has been an integral part of the survey of the public lands since its inception.²⁸ As an illustration of this, the first survey

conducted under the rectangular survey system (Township 1, Range 1, Seven Ranges, Ohio, by Deputy Surveyor Absalom Martin) included meandering of the Ohio River. Typical of early instructions for the Public Land Survey, the 1842 instructions to deputy surveyors in Florida addressed meandering as follows:

You will accurately meander, by course and distance, all navigable rivers which may bound or pass through your district; all navigable bayous flowing from or into such rivers; all takes and deep ponds of sufficient magnitude . . .

The first manual to provide detailed instructions for the location of the meander lines was the 1919 *Manual of Instruction*. That manual specifically called for the meander line to be run at “mean high water elevation” and provided further detail describing the line using currently accepted definitions of the ordinary high-water mark. In tidal waters, the instructions obviously will not yield as precise and repeatable a line as a more sophisticated mean high-water line determined by modern techniques. Nevertheless, in most cases, the line found by such instructions would be a good approximation of the mean high-water line. Therefore, it appears that the meander line was at least intended to be, and should be in most cases, a reasonable approximation of the ordinary high-water line or mean high-water line at the time of the survey.

Despite the fact that meander lines were intended to approximate the mean high-water line in tidal waters or the ordinary high-water line in nontidal waters, the location of a meander line generally may not be used as a boundary. This is due to the fact that the body of water itself is the natural monument and the current boundary when called in a description, not the representation provided by the meander line. Rather, the meander line represents only an approximation of the location of the boundary at a point in time. This concept is reflected in a number of court opinions typified by the following:

It is an established and accepted principle, subject only to the exception hereinafter noted, that the meander line of an official government survey does not constitute a boundary, rather the body of water whose shoreline is meandered is the true boundary . . .²⁹

Nevertheless, there may be instances where the current water boundary cannot be located and the meander line may be accepted as the boundary by default, or where the discrepancy between the meander line and the shoreline is large enough to indicate intentional omission of certain lands or fraud. These instructions are illustrated by the following exceptions from court opinions:

Under the circumstances of this case [Testimony had indicated that the true mean high-water line could not be located with any certainty], we hold that the meander line constituted the boundary line between the swamp and overflowed lands and the sovereignty lands . . .³⁰

However, a meander line may constitute a boundary where so intended or where the discrepancies between the meander line and the ordinary high-water line leave an excess of unsurveyed land so great as to clearly and palpably indicate fraud or mistake.³¹

In addition, the meander line has value as a representation of the location of the boundary at a certain point in time in the process of apportionment of newly formed land created by accretion and reliction. Therefore, although certain limitations must be recognized, the location of a meander line can have legal significance for boundary determination.

9.9 OWNERSHIP OF NON-PUBLICLY OWNED SUBMERGED LANDS

Principle 9. *Submerged lands under non-publicly owned waters are customarily owned by the adjoining owners, with the ownership extending to the center of the stream.*

Title to the submerged lands underlying non-public trust waters generally is in private ownership. As a result, boundary questions in such waters usually involve the boundary between two private owners.

Perhaps the most elementary example of this type of boundary is that involving a stream as the boundary between two parcels of land. In such a case, when the deeds of either side call “to the stream,” the center of the stream would be the boundary. There is general agreement on the part of most courts on this issue. Yet there are two complicating issues relating to stream boundaries in which there appears to be a divergence among court opinions. These issues deal with the questions of what constitutes the “center” of the stream and how the lateral boundaries between the upland boundaries and the center of the stream should be run.

Regarding the center of the stream, one approach defines the boundary as the “thread of the stream,” which is defined as the line lying equidistant between the banks. For example, one source,³² supported by a number of court cases, states as follows:

The term “thread of the stream” means the geographic center of the stream at ordinary or medium stage of the water, disregarding slight and exceptional irregularities in the banks. It is fixed without regard to the main channel of the stream. . . . If the stream is made a boundary in a private conveyance, then the thread of the stream will be the stream boundary.

The second approach regarding the center of the stream defines the boundary as the *thalweg*, or deepest part of the channel. As an example of this approach, *Stubblefield v. Osborn*³³ states as follows:

Upon principle, therefore, it would appear that the thread of a nonnavigable river is the line of water at its lowest stage. The thread or center of a channel, as the term is above employed, must be the line which would give to the landowners on either side access to the water, whatever its stage might be and particularly at its lowest stage.

There also seems to be more than one approach to how lateral boundaries are extended to the center of the stream. At least one source³⁴ indicates that straight-line extension of the upland boundary would be the correct approach for unmeandered (not meandered in a public land survey) streams. Other sources suggest that such

partition lines should be drawn perpendicular to the “line of navigation,” especially if the stream is sinuous, which would create inequitable boundaries with the straight-line approach. A third approach found in the *Manual of Instructions*³⁵ directs apportioning the length of a median line according to the length of a meander line of the shoreline of the adjacent riparian lots.

When nonnavigable lakes or ponds form the boundary line between two or more parcels, similar approaches are used as those for the proportion of riparian rights (see Section 9.14). For substantially round lakes, the general rule is that a center point is selected at the geographical center (or, arguably, the deepest point in the lake) of the lake, and partition lines are then run from the ends of the upland boundaries to the center point forming “pie slices.” For long or irregular lakes, the lines are typically run perpendicular to a centerline.

9.10 SWAMP AND OVERFLOWED LANDS

Principle 10. *Swamp and overflowed lands are not part of the public trust submerged lands. In the public land states, such lands were granted to the states for subsequent improvement and conveyance to private owners.*

Uplands that were unfit for agriculture in their natural state due to inundation were given a special classification in the U.S. Public Land Survey. They were not considered part of the navigable public trust submerged lands. However, since they were unfit for agriculture in their natural state, they were not considered to be suitable for conveyance to settlers without improvement. As a result, lands classified as such were granted to the individual states for improvement and resale based on a series of acts of Congress beginning in 1848. In some public land states, a large amount of lands was considered to fall in this classification and be conveyed to the states. For example, in Florida, over 20 million acres were conveyed to the state as swamp and overflowed lands out of a total area of the state of 35 million acres. The state, in turn, sold much of that amount to private entities for the purpose of reclamation.³⁶

For the purposes of classification, *swamplands* are those that “require drainage to make them fit for cultivation. *Overflowed lands* are those that are subject to such periodic or frequent overflows as to require levees or embankments to keep out the water and render them suitable for cultivation.”³⁷ Swamp and overflowed lands are not tidelands, nor are they part of the beds of lakes or nontidal streams. Rather, they are part of the floodplain lying above the public trust submerged lands. This classification is emphasized in the instructions to public land surveyors,³⁸ where it is stressed that meander lines, which approximate the upland boundary of public trust waters, should not be established at the segregation line between the upland and the swamp and overflowed lands but at the actual margin of the bed of the water bodies.

Once the United States divested itself from these “unusable” lands to the limited number of states, the states were free to dispose of them in any manner they desired. Some were drained, some were sold to individuals and corporations and some were retained by the states. In a recent case in Florida the state obtained swamp land that bordered a fractional section on the Atlantic Ocean. The original surveyors who

conducted the original township breakdown created a fractional Lot 5. The “water line” was surveyed and described in the field notes. The dividing line was listed in the field notes by courses and called a “delineation ” line between land and swamp. The State of Florida subsequently conveyed that portion of the fractional section that lay seaward from the fractional lot to an individual who in turn conveyed it to the railroad, who subsequently conveyed it to the Nature Conservancy who donated it to a federal wildlife refuge. After nearly 100 years the fractional lot was sold by the original patentee “according to the government survey.” The new owner reading the notes and with recommendations from his surveyor claimed water access and started to build a walkway. This claim was based on the presumption the “delineation line” was a meander line and thus the fractional lot extended to the ocean. The “delineation line” was visible to observation as a minutely raised ridge.

After a two day trial the court held that the surveyor has to take the surveyor’s notes literally. He did not call the line a meander line but a “demarcation” line; thus it was a boundary that was created. This intention was manifested when the federal government conveyed the land as a separate parcel and the Fractional lot as a separate parcel. In this instance the “paper trail” from the original notes to today’s deed became critical in explaining what a “demarcation line” was intended to be..

9.11 NAVIGATIONAL SERVITUDE

***Principle 11.** The United States has authority to regulate activities on all navigable waters of the country.*

The commerce clause of the U.S. Constitution confers on the United States a unique power of regulation and control over all navigable waters of the United States. For that purpose, they are the public property of the nation. This power extends to the entire stream and the streambed below the ordinary high-water mark. It allows the government to take property without compensation to private property owners. The proper exercise of this power is not an invasion of any private property rights in the stream or the lands underlying it, for the damage sustained does not result from taking property from riparian owners within the meaning of the Fifth Amendment but from the lawful exercise of a power to which riparian interests have always been subject. Thus, without being obligated constitutionally to pay compensation, the United States may change the course of a navigable stream or otherwise impair or destroy a riparian owner’s access to navigable waters. It does not matter that the market value of the riparian owner’s land is substantially diminished.

9.12 PUBLIC REGULATION OF RIPARIAN AND LITTORAL LANDS

***Principle 12.** Riparian and littoral land is often subject to public regulation, which may create regulatory limit lines, which, although different from property lines, may also affect the utility of the property.*

Riparian and littoral property has traditionally been subject to certain restrictions in its use. The antiquity of this concept is evidenced even in the Institutes of Justinian, which, as indicated in the introduction to this chapter, required that privately owned riparian land remain open to the public, as follows:

The public use of the banks of a river is part of the law of nations, just as is that of the river itself. All persons, therefore, are as much at liberty to bring their vessels to the bank, to fasten ropes to the trees growing there, and to place any part of their cargo there, as to navigate the river itself. But the banks of the river are the property of those whose land they adjoin, and consequently the trees growing on them are also the property of the same person.³⁹

With increased population density in littoral and riparian areas since the time of the Roman Empire, there has been a commensurate requirement for increased mediation by governmental policy and increased restriction on land use in such areas. Examples of such regulation include coastal construction control, regulation of dredge and fill in riparian and littoral wetlands, and regulation of development in floodplains.

Most coastal states have *coastal construction control lines* regulating or prohibiting construction in defined littoral areas. Typically, such lines are established by state agencies based on the height of the shoreline topography compared with the height of specific return frequency storm waves. Usually, the lines are monumented and have recorded metes and bounds descriptions.

Wetlands are natural areas where the groundwater table is normally at or close to the surface. Due to their value for plant and animal habitat, water purification, groundwater recharge, floodwater storage and flood peak reduction, and carbon storage, many states as well as the federal government regulate dredge and fill and construction in riparian wetlands. Typically, jurisdiction lines delimiting regulated wetlands are established only on an as-needed basis. Such wetlands are usually defined by the predominance of certain indicator species of vegetation, the presence of hydric soils, and/or the frequent presence of groundwater near the surface.⁴⁰

Floodplains are natural overflowed areas adjoining surface waters. While some floodplains are also wetlands, many are not, even though they are subject to flooding during storm events. Floodplains are often defined based on the probability of being flooded in any given year. The primary concerns regarding land use in floodplains are the loss of life and property from flooding that can result when houses are constructed in these areas, together with concern that alteration of the natural topography of the floodplain could cause worse flooding. Activity in such areas is often regulated on the local, state, and federal level. Federal floodplain regulation is administered through the National Flood Insurance Program administered by the Federal Emergency Management Agency. Generally, regulations regarding floodplains allow agriculture and similar low-impact use as long as no topographical alterations are made. Construction is usually not allowed in the floodway zones closest to the surface water but is generally allowed in the outer portions of the floodplain with the provision that occupied areas and such facilities as septic tanks be elevated above maximum flood elevations.

In the foregoing examples of regulation of riparian and littoral lands, the land generally remains in the ownership of the upland owner, yet the use of the land may be severely restricted. In such cases, the regulation line often takes on a greater significance to the use of the land than that of the actual water boundary. As a result, it often becomes necessary to locate pertinent regulation lines in addition to or in lieu of the water boundary itself. Therefore, research for the location of riparian and littoral boundaries typically includes investigation into pertinent land use regulations.

9.13 SHORELINE CHANGES AND WATER BOUNDARIES

Principle 13. *The general rule is that water boundaries change with gradual and natural changes to the shoreline, while those boundaries do not change with sudden or human-made changes. Land that is gradually and naturally built up by accretion or exposed by reliction accrues to the riparian owner. Land that is gradually and naturally eroded away or permanently inundated by the water accrues to the owner of the bed.*

Water boundaries generally change with and follow changes in the position of the shoreline. Shoreline changes may be categorized as those relating to changes in water level including withdrawal of the water (*reliction*) and rises in water level; and changes in land form including deposition of material along the shoreline by water action (*accretion*) and the wearing away of the land by water and wind action (*erosion*).

The general rule is that the upland owner gains title to new upland created by reliction and accretion, and loses title to lands submerged by rises in water level or lost by erosion. The general rule would presumably also apply to new upland created by the rising of land due to glacial rebound. That phenomenon is common in areas near melting glaciers. An exception to the general rule may occur when the shoreline changes are not gradual or imperceptible, or where such changes are artificially induced. Where such changes occur suddenly, such as during storms, this is termed *avulsion*, and it is generally held that title does not change with such shoreline changes. Likewise, it is usually held that shoreline changes resulting from human-made actions, such those associated with dredging or construction of groins, do not change title if the upland owner or a predecessor in title caused the changes.

There are exceptions to these general rules. For example, in regard to the ownership of accreted lands, it has been held in Louisiana that accretions along the Gulf of Mexico (but not along streams) belong to the state. A similar exception exists in the state of Washington.

Regarding erosion and inundation, land gradually eroded away or permanently inundated by water becomes part of the bed of the water body and generally belongs to the owner of the bed. In sovereign-owned water bodies, the ownership changes from the riparian owner to the public as owners of the bed. However, if a riparian owner already owns the bed or owns to the middle of the water course, ownership remains the same.

Unlike gradual erosion, land is not lost by avulsive action. Therefore, land detached from the riparian land by the sudden process of avulsion belongs to the owner from whose land it was attached. Similarly, land created by sudden action does not accrue to the ownership to which it is attached. A river that changes its course by cutting a new channel and forming an island does not change the ownership of the new island. Similarly, ownership of a parcel of ground detached from the land of one owner and attached to the land of another by the sudden change of a river course is not lost by the original owner. In some states, a statute of limitation determines how soon the former owner must claim the detached land. For example, in some states, when a riparian owner on a publicly owned water body loses land due to avulsion, the riparian owner has the right to fill the newly submerged lands if accomplished within a certain period of time after the loss.

Human-caused shoreline changes are generally referred to as *artificial changes*. Such changes may alter a shoreline directly, as by installation of fill or a jetty, or may indirectly cause artificial accretion that changes the shoreline. Generally, direct artificial shoreline changes do not change title. As a general rule, the owner of the bed of submerged land becomes the owner of fill placed on it. But land created at the water's edge as the result of filling or dredging by a third person has been held or recognized to be the property of the adjacent upland owner as with natural accretion. It has been generally held that shoreline changes resulting from human-made actions, such as those associated with dredging or groins, do not change title if the upland owner or a predecessor in title to the upland owner caused the changes, but may change title if the shoreline changes were caused by action beyond the control of the upland owner. As an illustration of this, a Florida case⁴¹ held that artificial accretion caused by the upland owner remains the property of the sovereign, while another case in that state⁴² held that artificial accretion caused by third parties will accrue to the upland owner.

When erosion or permanent inundation causes all of a riparian tract to be submerged, the new land touched by the water becomes riparian. If the original riparian land reappears due to accretion or reliction, judicial guidance appears to vary as to whether the reappeared land belongs to the original owner or to the newly riparian owner. As an example, in *Yearsley v. Gipple*,⁴³ the court held that if lands become riparian by the eroding away of adjoining lands, the owner is entitled to the right of the riparian owner to accretions, even though they extend beyond the original boundary line of the land. Conversely, in *Allard v. Curren*⁴⁴, land in Section 30 was eroded away by the Missouri River until land in Section 31 became riparian. By accretion, land in Section 30 was restored. The restored land in Section 30 was ruled not to be the property of the owner of Section 31. A consideration in such decisions appears to be whether the submergence was long continued so as to preclude identity upon reliction and whether erosion transported soil beyond the limits of ownership.

Based on all of this, it may be seen that the ambulatory nature of riparian boundaries requires the use of additional procedures to those required for the more conventional land boundaries. The first of these is that it is necessary to associate a *time* with the determination of a water boundary. The location today may not be the same as the location of that boundary a few years later. A second requirement is the need

to determine *the nature and causes of any changes* that have taken place, considering the foregoing discussions on avulsion and human-caused changes. A third requirement is the necessity at times to determine the *location of historic shorelines* when shoreline changes have occurred that have not resulted in a change of title.⁴⁵

9.14 APPORTIONMENT OF RIPARIAN AND LITTORAL RIGHTS

Principle 14. *Regarding the apportionment of riparian and littoral rights:*

- *The objective of apportionment is to distribute either newly formed land or rights in water in proportion to the length of water frontage. Therefore, the length of the affected tract's water frontage, not its total area or upland lateral lot lines, generally controls the apportionment of riparian rights.*
- *For the division of newly formed land, the point of departure for a division line is the intersection of the upland lateral boundary and the shoreline at the time of the subdivision of the area. For the division of water area, the point of departure for a division line is the intersection of the upland lateral boundary and the current shoreline.*
- *For the division of newly formed land, the newly formed shoreline should be proportioned relative to the respective lengths of the old and new shoreline.*
- *In streams, the general rule for the division of water area is to project the division line in a direction perpendicular to the thread of the stream.*
- *In large bays, oceans, or wide rivers where a thread is not readily discernible, the general rule for the division of water area is to run lines perpendicular to various other baselines such as the shoreline if it is relatively straight, the bulkhead or pier line, or the channel line.*
- *In lakes, the general rule for the division of water area for round lakes is for side lines to be drawn to a center point, creating pie slice-shaped areas. For long lakes, it is generally held that a centerline should be constructed and lines projected out perpendicular to the centerline.*
- *In partially enclosed areas, such as coves, the general rule for the division of water area is to proportion the closing line across the mouth of the area based on the respective lengths of the shoreline.*

Riparian and littoral rights are those rights associated with ownership of land fronting on water. Actually, riparian rights (derived from the Latin *ripa*, “a river-bank”) apply only to lands bordering rivers and streams, while littoral rights (derived from the Latin *litus*, “the seashore”) applies to oceanfront lands. Due to the more general use of the word, *riparian* will be used in this writing for rights associated with all types of water bodies. These rights include the right to newly formed upland created by accretion or reliction. In addition, these rights also may include a wide range of privileges, subject to state law, associated with the use and enjoyment of waterfront property, including reasonable use of water for domestic, irrigation,

and livestock use, construction of docks and mills, access to navigable channels, and scenic views across the water.

While the method for determining limits of the riparian rights area of a tract may vary significantly with the situation, the objective is to apportion either the newly formed land or the water area over which riparian rights are available in proportion to the relative length of each riparian tract. As a result, the length of the affected tract's water frontage, not its total area or the direction of the upland lateral lot lines, generally controls the apportionment of riparian rights.

Where the limits of rights to newly formed land are at issue, it is generally held that the starting points for lines dividing the new land among the affected riparian owners is at the intersection of the lateral upland lot line and the shoreline at the time of the subdivision of the area. For the division of water areas for other riparian rights, the starting points for the division lines are at the intersection of the lateral upland lot lines and the current shoreline, such as the mean high-water line or the ordinary high-water line.

For the division of newly formed land, it has been generally held that the newly formed shoreline should be proportioned relative to the respective lengths of the old shoreline (Figure 9.1). This is usually accomplished by measurement of the newly formed shoreline and use of the original shoreline from the survey subdividing the area.

For the division of water area, there are typically not two shorelines to proportion, as with the division of newly formed land. Therefore, other means must be used to provide comparable results. The method used for this purpose varies with the nature of the water body. It has been generally held that the proper procedure for this does *not* involve extension of the upland boundary without change of direction. Therefore, the direction of the upland boundaries are generally not controlling for division lines for riparian rights.

In rivers that are significantly narrow for determining a *thread* or center of stream, the procedure suggested in most cases is to project the line in a direction perpendicular to the thread of the stream. The "thread" of a stream is the line equidistant from the two banks of the stream. Typical cases suggesting this approach include *Campau Realty Co. v. Detroit*.⁴⁶ Where the geometric center of the stream is considerably removed from the thalweg, or deepest part (channel) of the stream, there may be justification for using the thalweg as the center. Location of the thalweg requires the use of hydrographic soundings.

In bays or oceans, as well as in wide rivers where a thread is not readily discernible, case law generally suggests using lines run perpendicular to various other baselines such as the shoreline if it is relatively straight, the bulkhead or pier line, or the channel line. Among numerous other cases, this approach has been specifically held in the Florida case of *Hayes v. Bowman*,⁴⁷ which dealt with a wide water body, where the court held that lateral lines run normal to the channel line would be the proper division lines (Figure 9.2).

One use of this approach is in the State of Mississippi, where riparian rights are limited to 750 yards waterward of the mean low-water line by statute. Therefore, in that state, for rivers and other water bodies greater than 1500 yards in width, the lateral lines for the riparian rights areas are either run perpendicular to a line offset

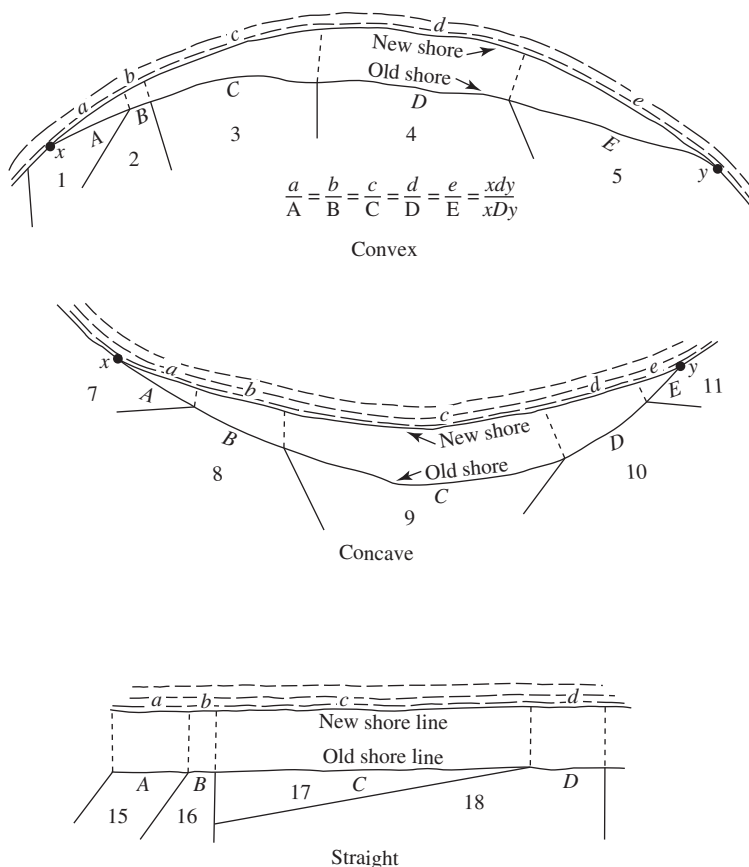


Figure 9.1 Proportionate shoreline method for distributing newly formed land.

750 yards from the mean low-water line or the area can be proportioned based on the respective lengths of the shoreline and the offset line.⁴⁸

Where the shoreline in question is curving or irregular such as in coves, the general direction from case law is to give each riparian tract a proportionate share based on measurement of the shoreline in question as well as measurement of some outer line such as the thread of the stream or the closing line of the cove.⁴⁹

Where riparian rights on lakes are involved, different approaches often are required, since a closed figure is involved. In round lakes, most case law has suggested side lines drawn to a center point, creating pie slice-shaped areas (Figure 9.3) such as suggested in *Hansen v. Rice*.⁵⁰ For long lakes, the judicial direction has been to use an approach similar to that used for rivers with a thread or centerline being constructed and lines projected out perpendicular to the center line (Figure 9.4), as suggested in *Rooney v. County of Stearns*.⁵¹ Another approach for long lakes with

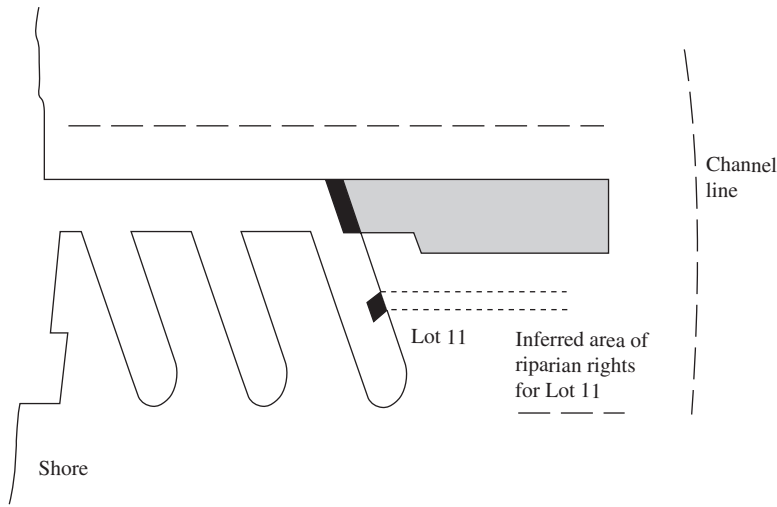


Figure 9.2 Riparian rights area determination with wide water body (*Hayes v. Bowman*).

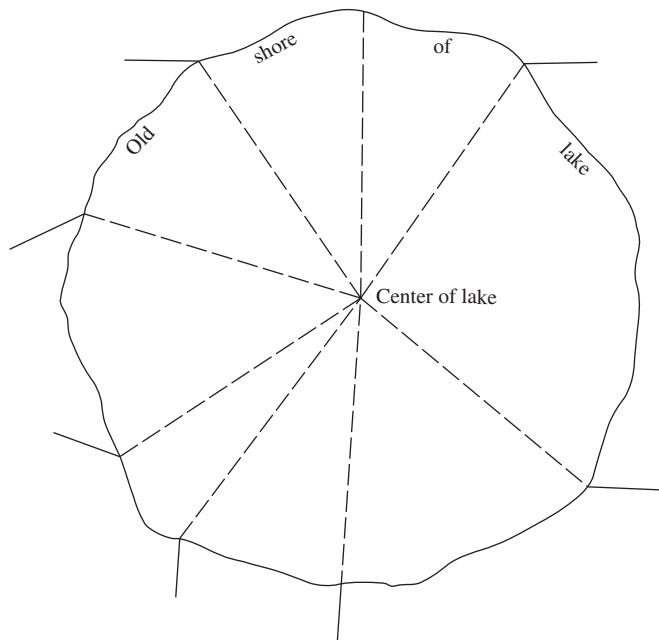


Figure 9.3 Round lake method for proportioning riparian rights.

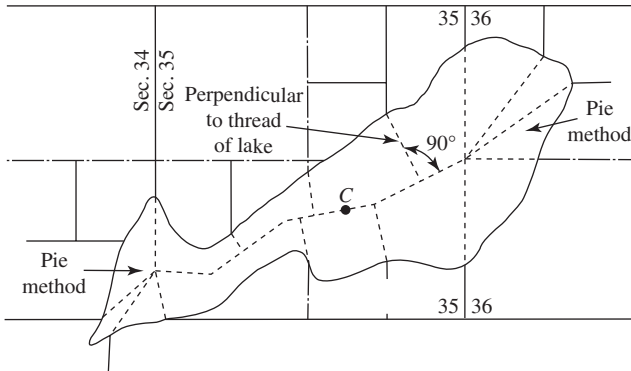


Figure 9.4 Long lake method for proportioning riparian lakes.

straight shorelines is to use division lines run perpendicular to the shoreline. For large, irregularly shaped lakes, proportionate division of the line of navigable water, similar to that suggested for irregular shorelines in rivers, has been suggested.

A proportionality test may be used to evaluate the fairness of a proposed apportionment. This technique compares the ratio of the areas allotted to two adjacent riparian lots to the ratios of the lengths of the shorelines of the two lots. Such an evaluation is based on methods used to evaluate proposed division of continental shelf areas for international boundaries.⁵² If the ratios are not similar, the proposed division may be considered to be inequitable.

Reviewing all of this, it may be seen that there are at least four basic cases—narrow streams, wide water bodies, irregular shorelines, and lakes—with each case having its own rules. With narrow rivers, the general rule is to use a division line perpendicular to the thread of the stream. With wider water bodies, the general rule is to use a line run perpendicular to the shoreline, offset line, or to some other baseline such as a bulkhead or channel line. For most lakes, the general rule is to use lines drawn to a common center point or centerline. For curving or irregular shorelines, proportional division of the centerline based on length of the shoreline is the general rule.

Because of variations in case law, there are other approaches for certain applications such as the Colonial Method used for division of tidal flats in some of the New England states (Figure 9.5). That method uses proportionate measurement between the low-water and high-water mark. Instructions for the use of that method as prescribed in the case of *Emerson v. Taylor*⁵³ are as follows: “Draw a base line from the two corners of each lot, where they strike the shore, and from those two corners, extend parallel lines to low-water mark, at right angles with the base line. If the line of the shore be straight there will be no interference in running the parallel lines. If the flats lie in a cove, of a regular or irregular curvature, there will be an interference in running such lines, and the loss occasioned by it must be equally borne or gain enjoyed equally by the contiguous owners. . . .”

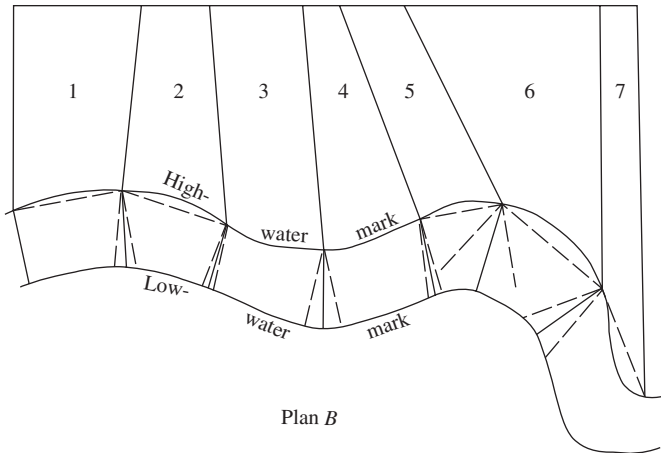


Figure 9.5 Colonial Method for proportioning tidal flats.

9.15 EMERGENT OR OMITTED ISLANDS

Principle 15. *Emergent islands are generally considered to belong to the owner of the bed of the water body. In public land states, omitted lands (such as islands existing at the time of, but not surveyed in, the public land survey) remain in the ownership of the federal government until specifically conveyed. Islands emerging after the conveyance of the water body to the states with statehood, belong to the owner of the bed of the water body.*

The general rule regarding ownership of accreted lands also applies to emergent islands, where it is generally held that islands formed in a water body belong to the owner of the bed of water body. If an island forms on one side of a stream and the riparian proprietors own the bed of the stream, the island belongs to the proprietor on whose side of the stream it formed. If an island is formed in the middle of a stream and the riparian proprietors own the bed of the stream, the island is divided by the line through the thread of the stream. When the public of the state owns the bed of the stream or lake, the adjoining riparian owners are not entitled to an island that emerges from the bed of the stream. The right to new islands follows the right of the soil on which they were formed. In public land states, islands that were existent at the time of the original public land survey but were overlooked in the survey are considered to be *omitted lands* that remain in the ownership of the federal government until validly conveyed by patent.

9.16 WATER BOUNDARIES OTHER THAN SEA

The International Boundary Research Unit at Durham University, England has estimated that today there are in excess of 2,000 boundary disputes dealing with water

boundaries between countries. Some countries have resorted to shooting while other have resorted to law. Many of these boundary issues are centuries old and the documentary evidence as well as physical evidence is scarce or not available. The United Nations and the World Court have attorneys and surveyors working worldwide trying to keep peace in the world.

At one or more times in their careers there are several classes of water boundaries that attorneys and surveyors should be acquainted with: Inland water boundaries of streams and lakes.

In dealing with these types of boundaries they are significant in that many are privately owned and landowners become very possessive of use by strangers.

Within this area of inland boundaries one can find meandered lakes, non-meandered lakes; navigable streams and rivers and non-navigable streams and rivers.

Streams and lakes become an important issue because many of the early surveyors used these natural features as boundary dividing lines between nations, states, counties, and individuals. One can see that the law has placed a high degree of legality of the accurate location of the features.

9.17 MAJOR RECOGNIZED AREAS

Some states are blessed with overabundance of natural lakes and rivers and some states have a scarcity. Each state has its own peculiarities. The natural divisions on inland water boundaries are meandered lakes of the GLO; non-meandered GLO lakes, private lakes, rivers as state boundaries; rivers as private boundaries. Private lakes all in one ownership; private lakes in multiple ownerships. Each and every one of these has different rules for surveying and for applying the laws.

9.18 CONCLUSIONS AND RECOMMENDATIONS

In this final section we wish to bring to the student and the professional who will or deal with water boundary issues, this area of boundaries is steeped with ambiguity and uncertainty. This area of boundary creation, retracement and litigation is still trying to find its way out of the “dark ages.” One can find a proliferation of recommendations on how water boundaries should be divided. None are absolute. After decades the courts are struggling with water issues. Our recommendation is to approach a water boundary problem with caution, but do an adequate job of research and do your professional best.

NOTES

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